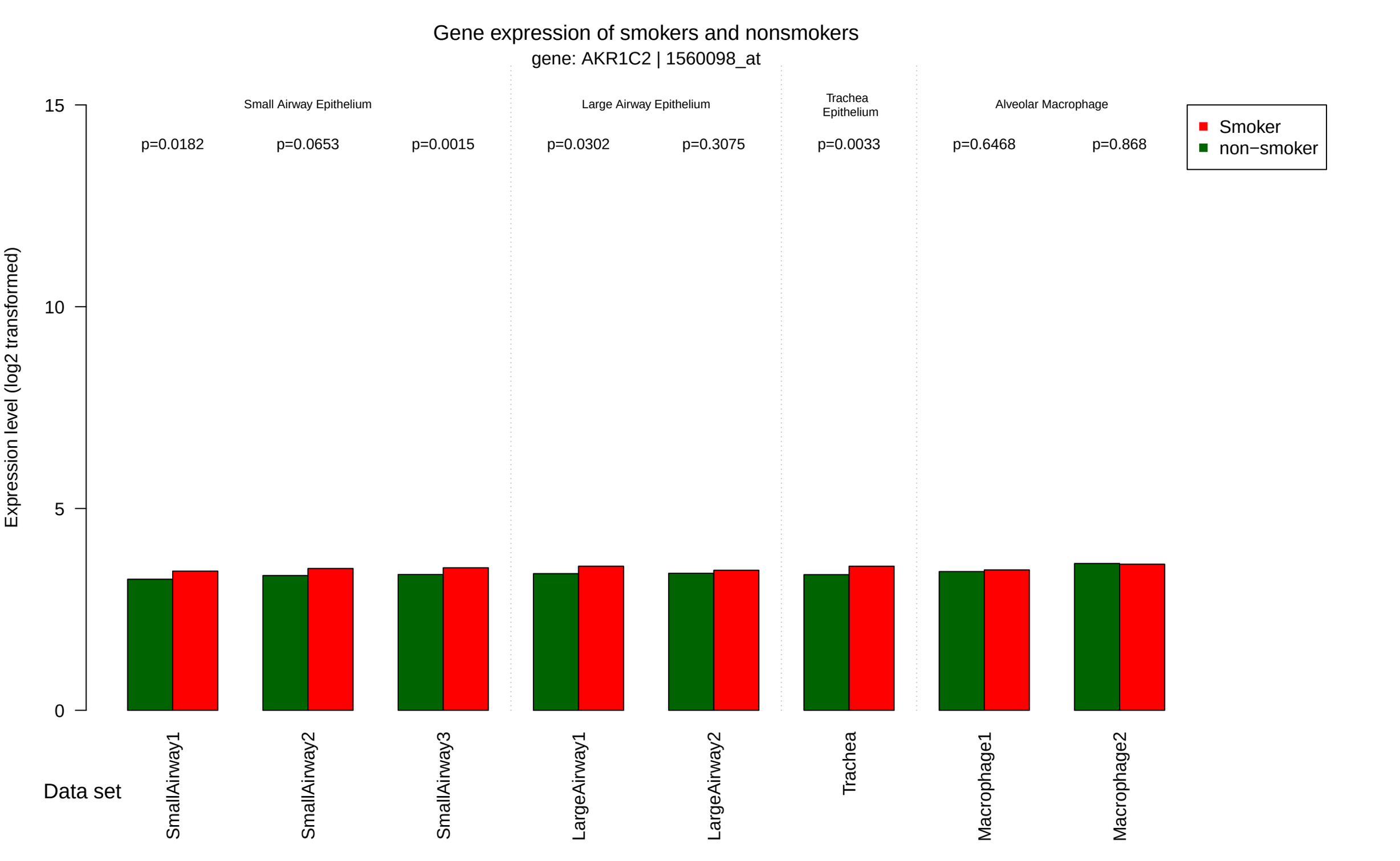
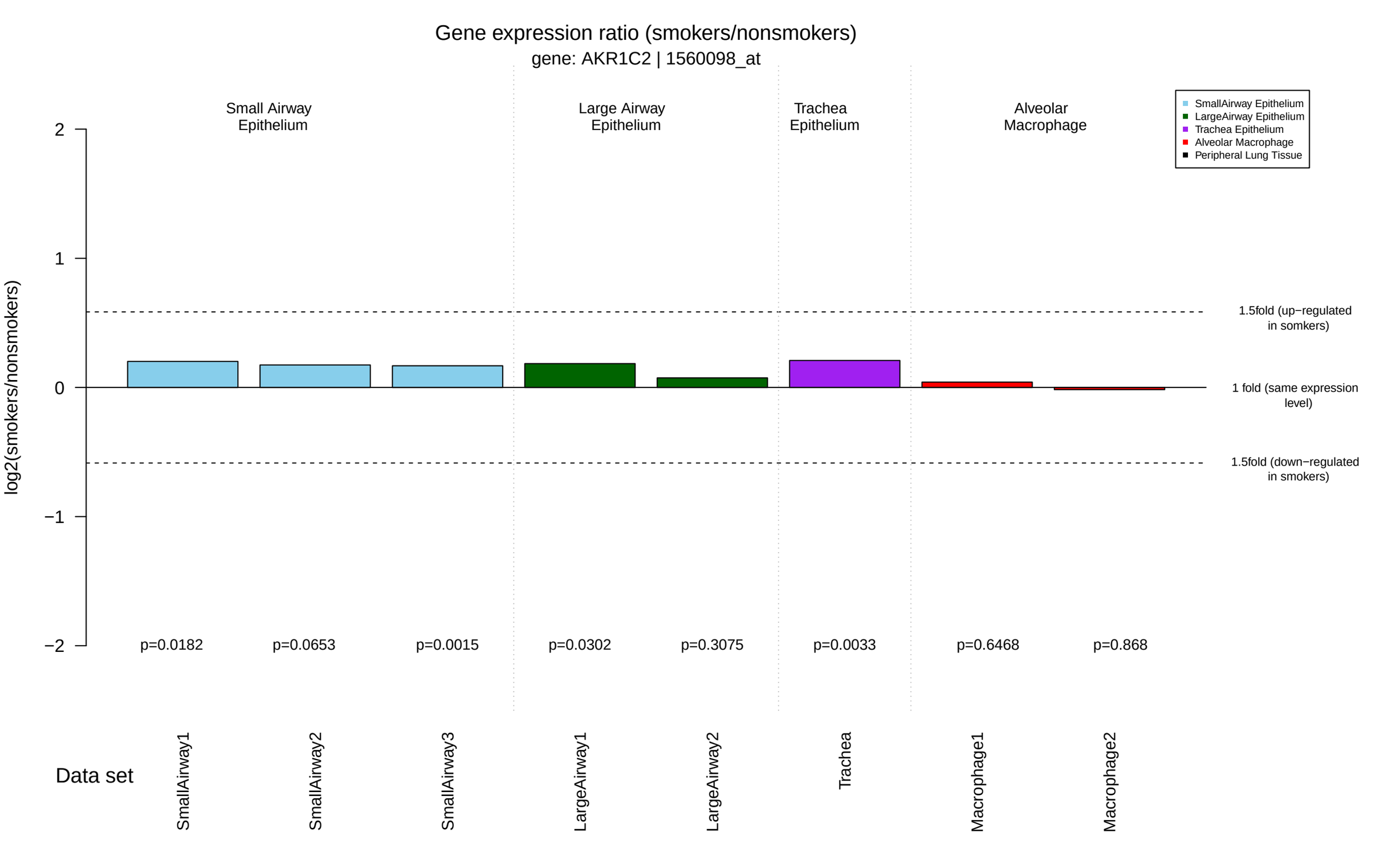
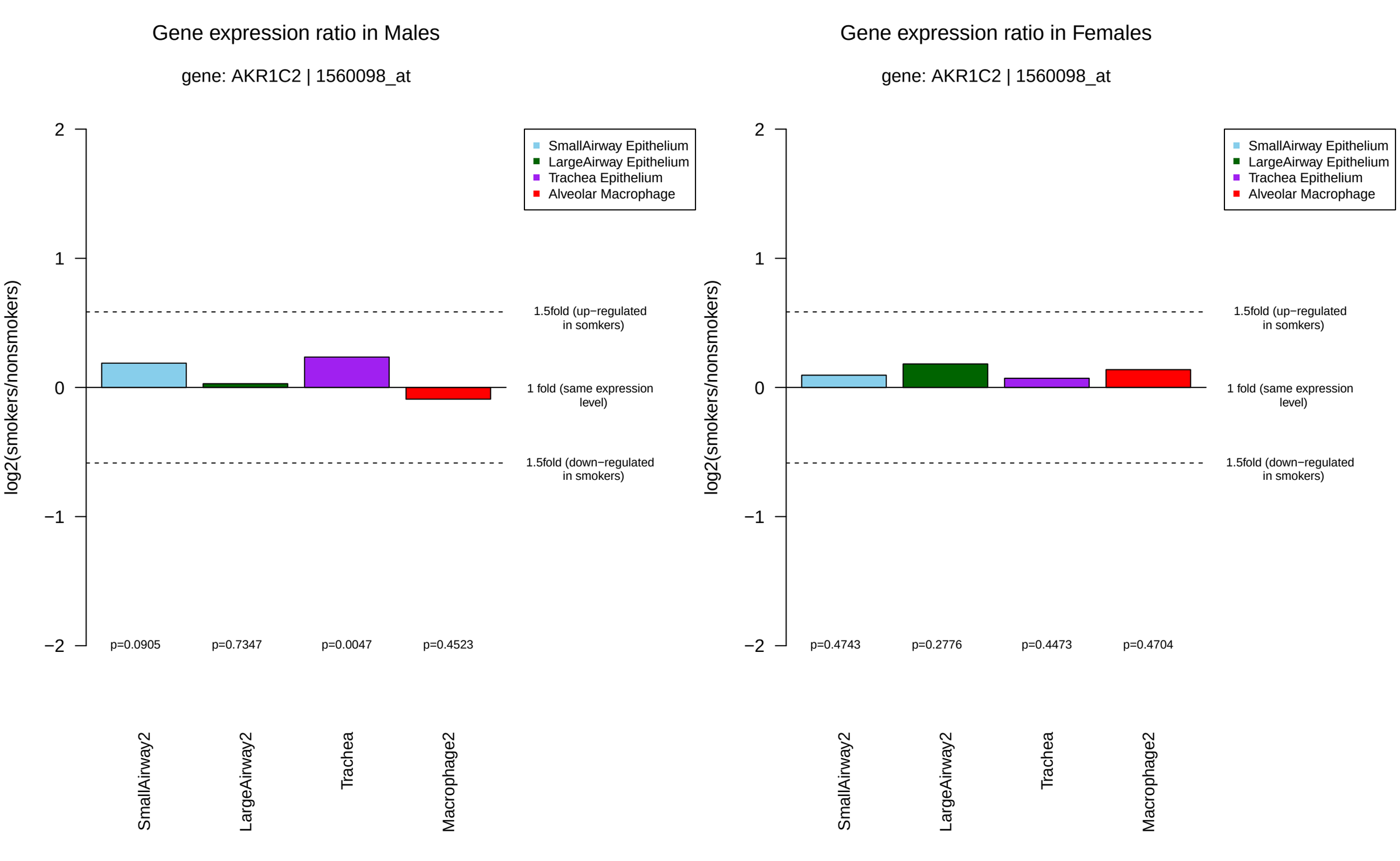
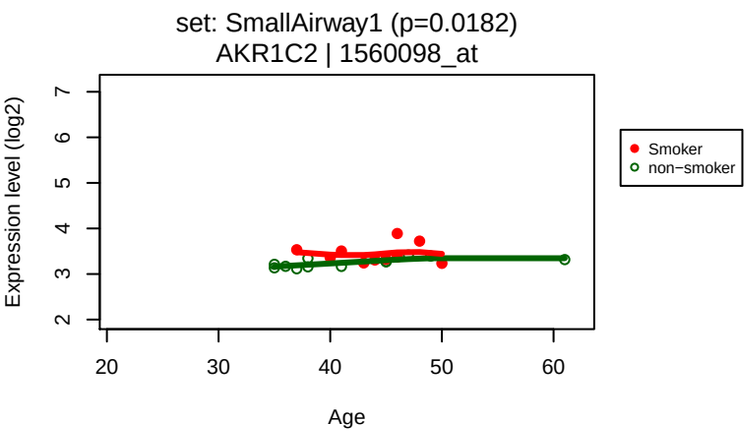
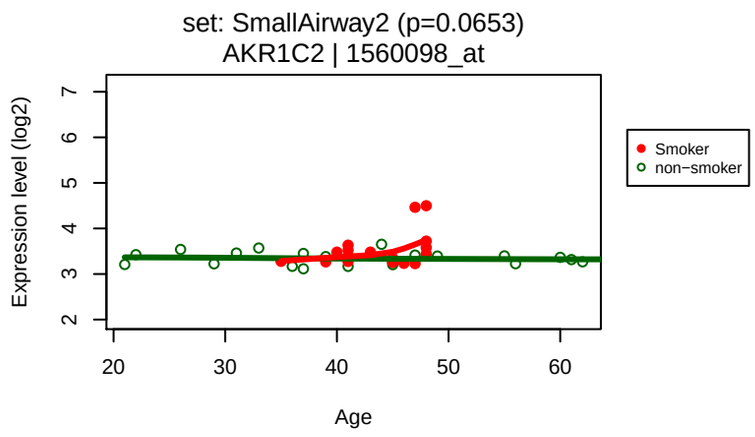
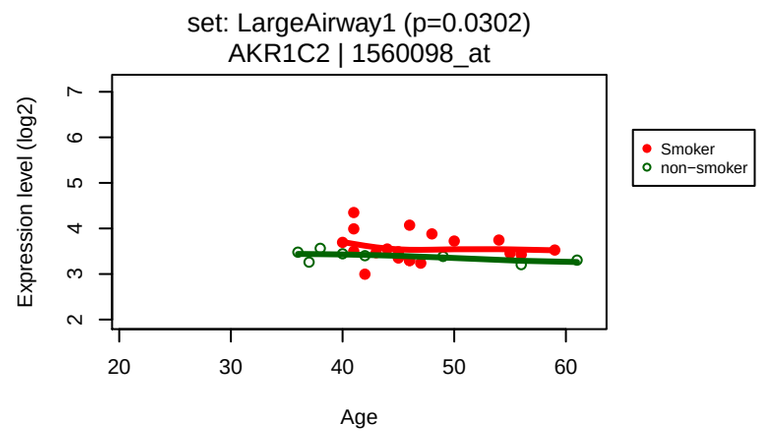

Gene expression of smokers and nonsmokers
gene: AKR1C2 | 1560098_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0182
p=0.0653
p=0.0015
p=0.0302
p=0.3075
p=0.0033
p=0.6468
p=0.868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

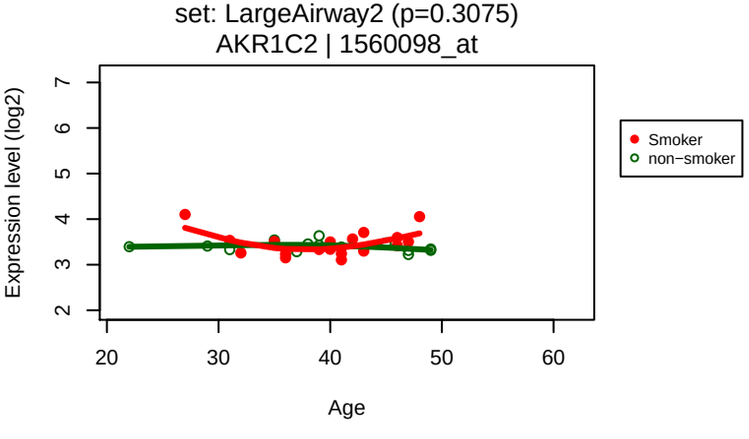

Gene expression ratio (smokers/nonsmokers)
gene: AKR1C2 | 1560098_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0182
p=0.0653
p=0.0015
p=0.0302
p=0.3075
p=0.0033
p=0.6468
p=0.868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

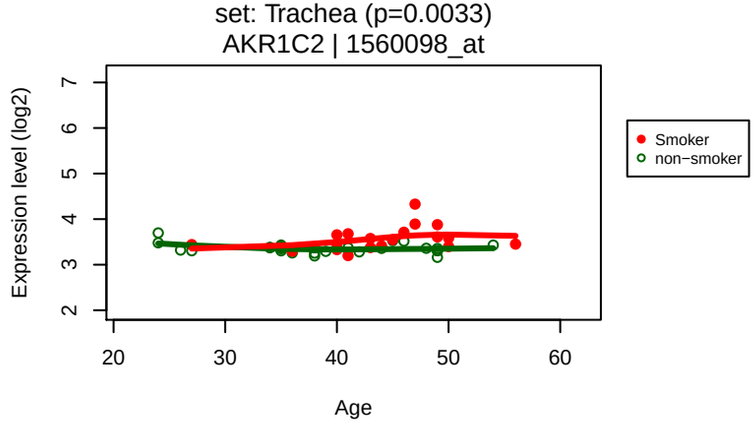

Gene expression ratio in Males
gene: AKR1C2 | 1560098_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0905
p=0.7347
p=0.0047
p=0.4523
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR1C2 | 1560098_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4743
p=0.2776
p=0.4473
p=0.4704
SmallAirway2
LargeAirway2
Trachea
Macrophage2

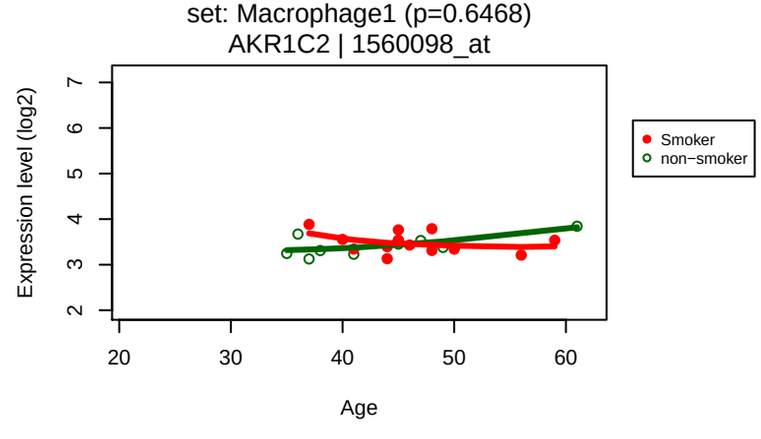

set: SmallAirway1 (p=0.0182)
AKR1C2 | 1560098_at
Expression level (log2)
Age
Smoker
non−smoker

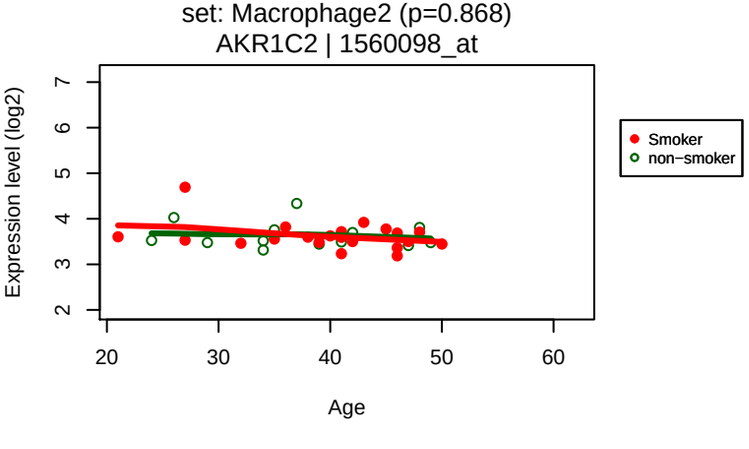

set: SmallAirway2 (p=0.0653)
AKR1C2 | 1560098_at
Expression level (log2)
Age
Smoker
non−smoker

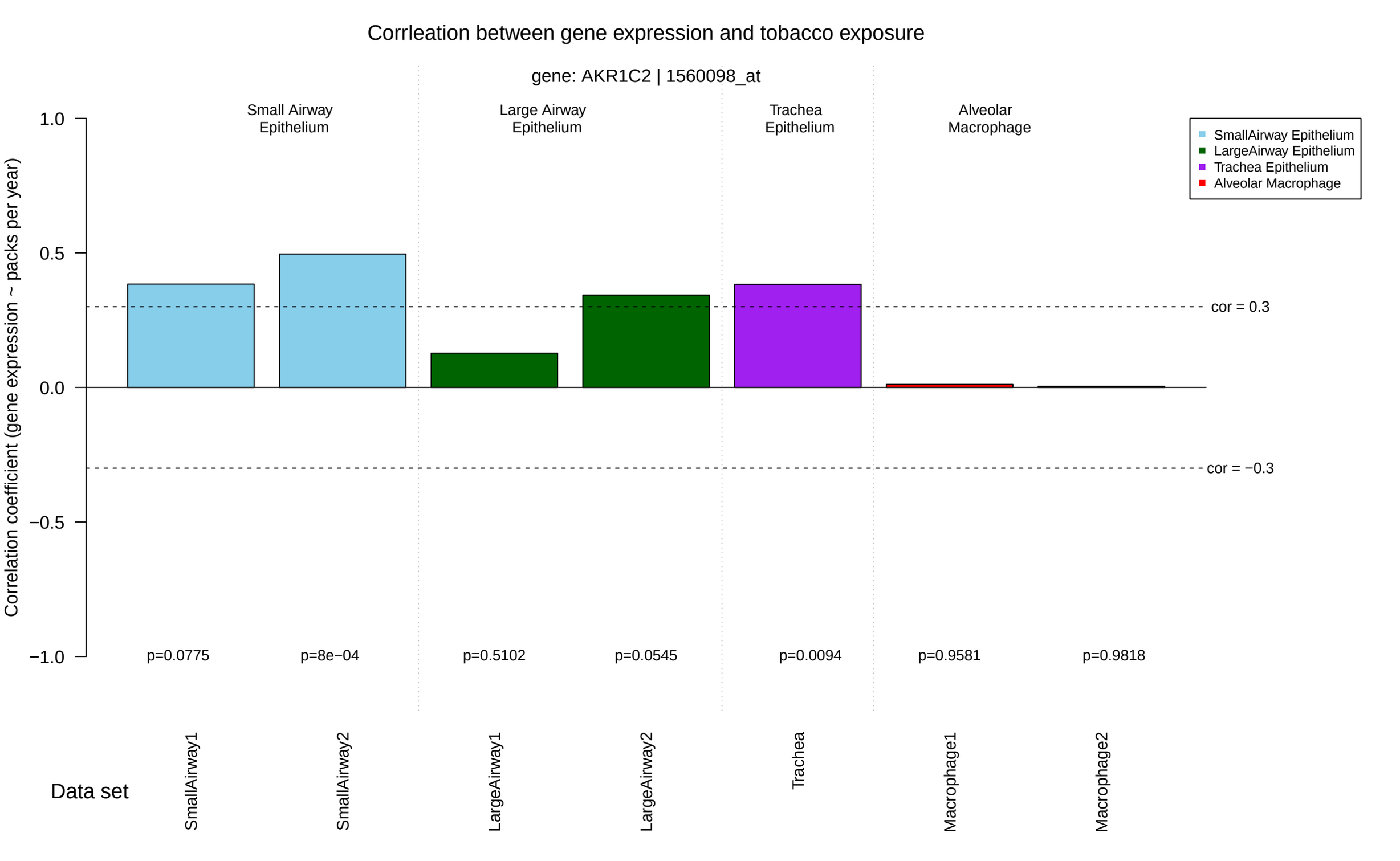

set: LargeAirway1 (p=0.0302)
AKR1C2 | 1560098_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3075)
AKR1C2 | 1560098_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0033)
AKR1C2 | 1560098_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6468)
AKR1C2 | 1560098_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.868)
AKR1C2 | 1560098_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR1C2 | 1560098_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0775
p=8e−04
p=0.5102
p=0.0545
p=0.0094
p=0.9581
p=0.9818
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2